

The Invertible Matrix Theorem

Let A be an $n \times n$ matrix. The following statements are equivalent:

1. A is invertible.
2. A is row equivalent to I_n .
3. A has n pivot columns.
4. $Ax = 0$ has only the trivial solution.
5. The columns of A are linearly independent.
6. The linear transformation $x \mapsto Ax$ is one-to-one.
7. The equation $Ax = b$ has a solution for all $b \in \mathbb{R}^n$.
8. The columns of A span $\mathbb{R}^n$.
9. The linear transformation $x \mapsto Ax$ is onto.
10. There is an $n \times n$ matrix C such that $CA = I_n$ (a left inverse).
11. There is an $n \times n$ matrix D such that $AD = I_n$ (a right inverse).
12. A^T is invertible.

LU Decomposition (Factorization)

Definition. A matrix $A = (a_{ij})$ is **upper triangular** if $a_{ij} = 0$ for all $i > j$. It is **lower triangular** if $a_{ij} = 0$ for all $i < j$.

Examples of upper triangular matrices:

$$\begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}, \quad \begin{bmatrix} 1 & -1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 3 \end{bmatrix}.$$

Examples of lower triangular matrices:

$$\begin{bmatrix} 0 & 0 \\ 2 & 1 \end{bmatrix}, \quad \begin{bmatrix} -1 & 0 & 0 \\ 2 & 3 & 0 \\ 1 & -1 & 0 \end{bmatrix}.$$

A matrix is both upper and lower triangular exactly when it is diagonal:

$$\begin{bmatrix} d_1 & 0 & 0 \\ 0 & d_2 & 0 \\ 0 & 0 & d_3 \end{bmatrix}.$$

Product of lower triangular matrices. If $L = (\ell_{ij})$ and $M = (m_{ij})$ are lower triangular, then for $i < j$,

$$(LM)_{ij} = \sum_k \ell_{ik} m_{kj} = 0.$$

Indeed, $\ell_{ik} = 0$ when $k > i$ and $m_{kj} = 0$ when $k < j$, so no term can be nonzero when $i < j$.

Thus, the product of lower triangular matrices is lower triangular. Similarly, the product of upper triangular matrices is upper triangular.

Definition. A **permutation matrix** is an $n \times n$ matrix P such that

$$P \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} x_{i_1} \\ \vdots \\ x_{i_n} \end{bmatrix}, \quad \{i_1, \dots, i_n\} = \{1, \dots, n\}.$$

Examples:

$$P \begin{bmatrix} 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The 2×2 permutation matrices are

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Example.

$$P \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} x_2 \\ x_3 \\ x_1 \end{bmatrix}, \quad P = [P(e_1) \ P(e_2) \ P(e_3)] = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}.$$

Theorem (Permuted LU Factorization). If A is an $m \times n$ matrix, then

$$A = PLU,$$

where P is an $m \times m$ permutation matrix, L is lower triangular with 1s on the diagonal, and U is a row echelon form of A .

In an intermediate step from A to RREF, pivots are nonzero; entries marked * may be any number.

Motivation for LU factorization. Suppose we are solving the systems

$$Ax = b_1, \quad Ax = b_2, \quad \dots, \quad Ax = b_k$$

for one matrix A and different vectors b_i . Repeating row operations on A while reducing $[A \mid b_i]$ to RREF repeats work that can be reused.

Use of LU factorization. Given $A = PLU$, solve $Ax = b$ by splitting it into two systems:

$$Ux = y, \quad Ly = P^{-1}b,$$

where P^{-1} is the inverse permutation matrix. Since $b = Ax = PLUx$, let $y = Ux$; then $Ly = P^{-1}b$.

Example.

$$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}.$$

Thus $x_2 = \frac{1}{3}$ and $x_1 = \frac{1}{3}$.

If A is $n \times n$, then finding $A = PLU$ takes $O(n^3)$ flops.

A **flop** is an operation in floating-point arithmetic. $O(n^3)$ means a constant times n^3 .

Solving $Ax = b$ takes $O(n^3)$ flops, while solving the triangular systems $Ux = y$ and $Ly = P^{-1}b$ takes $O(n^2)$ flops.

Simplifying assumption. Suppose A can be reduced to REF (or equivalently to RREF) without row swaps.

Algorithm

1. Reduce A to REF using row replacement only (no scaling).
2. Record the row replacements in L .

Example.

$$\begin{aligned} A &= \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & -1 \\ -1 & 1 & 2 \end{bmatrix} \\ &\sim \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 3 & 1 \end{bmatrix} \\ &\sim \underbrace{\begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 4 \end{bmatrix}}_U. \end{aligned}$$

The record of row reduction from A to U is

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix}.$$

Indeed,

$$LU = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & -1 \\ 0 & -1 & 1 \\ 0 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & -1 \\ -1 & 1 & 2 \end{bmatrix} = A.$$

If A reduces to REF U using only row replacements, then

$$E_k \cdots E_1 A = U,$$

where the E_i are elementary matrices for the row replacements. They are lower triangular because each operation adds a multiple of an upper row to a row below it. Therefore,

$$A = \underbrace{E_1^{-1} \cdots E_k^{-1}}_L U.$$

Each E_i^{-1} is also lower triangular, and their product is lower triangular.

Subspaces

Definition. A set of vectors $S \subseteq \mathbb{R}^n$ is a **subspace** if S is closed under addition and scaling:

1. If $u \in S$ and $c \in \mathbb{R}$, then $cu \in S$.
2. If $u, v \in S$, then $u + v \in S$.

Theorem. S is a subspace if and only if, for any $c_1, \dots, c_k \in \mathbb{R}$ and $u_1, \dots, u_k \in S$,

$$c_1u_1 + \dots + c_ku_k \in S.$$

Example. The solution space of $Ax = 0$ is a subspace. If $Ax = 0$ and $Ay = 0$, then

$$A(cx) = c(Ax) = c \cdot 0 = 0,$$

and

$$A(x + y) = Ax + Ay = 0 + 0 = 0.$$

Definition.

$$\text{Nul } A = \{x \in \mathbb{R}^n : Ax = 0\}.$$

If $T(x) = Ax$, then

$$\ker T = \text{Nul } A.$$

Example. If $v_1, \dots, v_k \in \mathbb{R}^n$, then $\text{span}\{v_1, \dots, v_k\}$ is a subspace:

$$c(c_1v_1 + \dots + c_kv_k) = cc_1v_1 + \dots + cc_kv_k,$$

and

$$\begin{aligned}(c_1v_1 + \dots + c_kv_k) + (d_1v_1 + \dots + d_kv_k) \\ = (c_1 + d_1)v_1 + \dots + (c_k + d_k)v_k.\end{aligned}$$

Example. If $A = [a_1 \ \dots \ a_n]$ is an $m \times n$ matrix, then

$$\text{Col } A = \text{span}\{a_1, \dots, a_n\}$$

is a subspace of $\mathbb{R}^m$.

Every subspace contains 0, the zero vector.

Example. If $b \neq 0$, is $\{x \in \mathbb{R}^n : Ax = b\}$ a subspace?

No. $A0 = 0 \neq b$, so the set does not contain 0.

Example. Is

$$S = \left\{ \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \in \mathbb{R}^2 : x_1x_2 = 0 \right\}$$

a subspace?

No. The sum of two vectors on the coordinate axes need not lie on either axis.

Example. Is a line in $\mathbb{R}^2$ a subspace?

Yes, if and only if it passes through 0.

Example. Is $\{0\}$ a subspace of $\mathbb{R}^n$? Yes.

Definition. A **basis** of a subspace S is an independent set of vectors in S that spans S .

Example. The standard basis $\{e_1, \dots, e_n\}$ is a basis of $\mathbb{R}^n$.

Example. Find a basis for $\text{Nul } A$. Solve $Ax = 0$:

$$A = \begin{bmatrix} 1 & 2 & -1 & 1 \\ 2 & 3 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & -1 & 1 \\ 0 & -1 & 2 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 3 & -1 \\ 0 & 1 & -2 & 1 \end{bmatrix}.$$

Thus

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = x_3 \begin{bmatrix} -3 \\ 2 \\ 1 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 1 \\ -1 \\ 0 \\ 1 \end{bmatrix}.$$

The two vectors on the right span $\text{Nul } A$, are linearly independent, and form a basis for $\text{Nul } A$.